

Question number	Answer	Additional guidance	Mark
4(a)(i)	bind to an {oxygen (molecule) / O₂} (1)	Do not accept O Ignore binding to carbon dioxide	(1)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	An explanation that includes the following points: - must have R groups that are {polar / hydrophilic} (1) - so that the {haemoglobin / protein} {can dissolve in (red blood cell) cytoplasm / is soluble in water} (1)	Accept they are {polar / hydrophilic} Ignore dissolve in blood / plasma	(2)

Question number	Answer	Additional guidance	Mark
4(b)(i)	An explanation that includes two of the following points: - oxygen dissociation curve for {maternal / adult} Hb is shifted to the right of curve for fetal Hb (1) - because oxygen needs to diffuse from {maternal / adult} blood into fetal blood (1) - therefore fetal haemoglobin needs to have a higher affinity for oxygen (1)	Accept shifted down for shifted to the right Accept converse Accept converse Accept because oxygen needs to dissociate from maternal haemoglobin and bind to fetal haemoglobin	(2)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	An answer that includes the following points: - total number of amino acids in haemoglobin calculated (1) - percentage of amino acids that are different calculated (1)	574 / 287 14 / 13.6 / 13.59 Correct answer with no working (2)	(2)